

Exasol Prototype Challenge: Plant Maintenance Intelligence

Overview

A manufacturing customer wants to understand whether Exasol can help them move from raw machine telemetry to actionable maintenance intelligence.

They collect sensor data from machines across plants, such as temperature, vibration, pressure, runtime, power usage, operating mode, and error codes. Today, this data is mostly reviewed after issues occur. The customer wants a quick prototype that helps answer:

Which machines need attention soon, why, and what should the operations team do next?

This is intended to be a customer-demo-style prototype, not a production system or a perfect ML solution.

Task

Build a small working prototype using Exasol Community Edition that demonstrates how plant telemetry can be turned into useful maintenance-risk insights.

You are free to decide:

- data model
- mock data generation approach
- risk scoring logic
- UDF design
- queries/views
- presentation format
- You may generate mock data. The dataset does not need to be large; it only needs to be sufficient to demonstrate the concept clearly.
- The maintenance or ML score does not need to be highly accurate.
- We are more interested in how you structure the problem, make reasonable assumptions, and build something useful.

You are free to use any AI tool of your choice.

Please spend no more than 4–6 hours.

Outcome

Your prototype should be good enough to present in a 15-minute customer demo.

Why this is valuable to the customer

Explain how the prototype helps the customer move from raw telemetry to faster maintenance decisions, reduced downtime, better prioritization, or improved operational visibility.

Submission

Submit a GitHub repository or zip file containing the prototype. The repo should include a short document, such as README.md, that explains the solution clearly.

Please include:

1. All scripts/assets and setup/run instructions
 2. SQL queries/views used in the demo
 3. Sample output or screenshots if helpful
 4. Solution explanation
 5. Assumptions and trade-offs
-

Evaluation Criteria

We are not looking for a perfect solution. We are looking for how you think and build.